# CCNA – Module 4: Routing and Switching

## Assessment Code: M4-A1

**Program:** CCNA
**Module:** 4 – Routing and Switching
**Assessment Duration:** 3 Hours
**Platform:** Cisco IOS / Cisco Packet Tracer

---

---

# 📘 SECTION A — CONCEPT APPLICATION

## 💡 Question 1

### Scenario

You are the network administrator for a small startup with only two routers connecting two offices, and the network topology rarely changes.

### Question:

**Explain the operational differences between static and dynamic routing, and justify why configuring static routes would be the most efficient choice in this scenario.**

### Answer:

**Static routing** is a routing method where the network administrator manually configures routes on a router. These routes remain in the routing table until an administrator changes or removes them.

**Dynamic routing** uses routing protocols such as RIP or OSPF to automatically learn and exchange routes between routers.

| Feature | Static Routing | Dynamic Routing |
|---|---|---|
| Configuration | Manually configured | Learned automatically |
| Route updates | Manual | Automatic |
| Resource usage | Low | Higher |
| Complexity | Simple for small networks | Better for larger networks |
| Failure adaptation | Requires manual changes | Can automatically find alternate routes |
| Scalability | Limited | High |

For this startup, **static routing is the most efficient choice** because there are only two routers and the topology rarely changes. The administrator can easily configure and maintain the required routes manually. Using a dynamic routing protocol would introduce unnecessary configuration and protocol overhead for such a small and stable network.

**Conclusion:**
Static routing is suitable because the network is small, simple, and stable.

---

# 💡 Question 2

## Scenario

You are expanding the startup's network to five branch routers, and manually maintaining routes has become error-prone.

## Question:

**Compare RIP and OSPF as dynamic routing protocol options for this network, and recommend which one is better suited as the network grows, justifying your choice based on convergence time and scalability.**

## Answer:

**RIP (Routing Information Protocol)** and **OSPF (Open Shortest Path First)** are dynamic routing protocols, but they differ significantly in scalability and convergence.

| Feature | RIP | OSPF |
|---|---|---|
| Type | Distance-vector | Link-state |
| Metric | Hop count | Cost |
| Maximum hop count | 15 | No 15-hop limitation |
| Convergence | Slower | Faster |
| Scalability | Limited | High |
| Network size | Small networks | Medium to large networks |
| Path calculation | Bellman-Ford based | SPF algorithm |

For a network expanding to five branch routers and potentially growing further, **OSPF is the better choice**.

OSPF provides faster convergence and is more scalable than RIP. It can quickly react to topology changes and calculate suitable paths using the Shortest Path First algorithm.

**Conclusion:**
OSPF is recommended because it provides better convergence, scalability, and adaptability as the startup network grows.

---

# 💡 Question 3

## Scenario

You are troubleshooting two newly connected startup devices attempting to communicate for the first time through the same switch.

## Question:

**Explain how the switch populates its MAC address table and forwards frames between the two devices, including what happens before the table is populated.**

## Answer:

A switch learns MAC addresses by examining the **source MAC address** of incoming Ethernet frames.

For example:

1. Device A sends a frame to Device B.
2. The switch receives the frame on an interface.
3. The switch records Device A's source MAC address and the receiving port in its MAC address table.
4. If the destination MAC address is already known, the switch forwards the frame only through the appropriate port.
5. If the destination MAC address is unknown, the switch performs **unknown unicast flooding** and forwards the frame out of the relevant ports except the port on which it was received.
6. When Device B responds, the switch learns Device B's MAC address as well.
7. Future frames can then be forwarded directly between the two devices.

## Example

| Device | MAC Address | Switch Port |
|---|---|---|
| Device A | AA:AA:AA:AA:AA | Fa0/1 |
| Device B | BB:BB:BB:BB:BB | Fa0/2 |

The switch gradually builds its MAC address table by learning the source MAC addresses of incoming frames.

**Conclusion:**
Before the destination MAC address is learned, the switch floods the frame. Once the MAC address is learned, the switch forwards frames directly to the correct port.

---

# 💡 Question 4

## Scenario

You are designing the network for a startup with a Marketing team and an Engineering team that must be logically separated for security, even though both teams' devices connect to the same physical switch.

## Question:

**Explain the purpose of VLAN segmentation, and describe how an 802.1Q trunk link allows traffic for multiple VLANs to travel between the switch and the router over a single physical connection.**

## Answer:

A **VLAN (Virtual Local Area Network)** logically separates devices into different broadcast domains even when they are connected to the same physical switch.

For example:

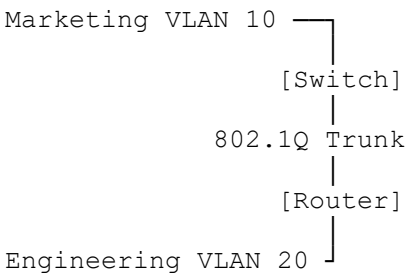
- VLAN 10 → Marketing
- VLAN 20 → Engineering

VLAN segmentation provides:

- Logical separation between departments
- Reduced broadcast domains
- Better network organization
- Improved security
- Easier network management

An **802.1Q trunk** allows multiple VLANs to travel over one physical link. The trunk uses VLAN tags to identify which VLAN each Ethernet frame belongs to.

For example:




The switch adds the appropriate VLAN information to frames travelling across the trunk. The router uses the VLAN information to process traffic for the correct VLAN.

**Conclusion:**
VLANs provide logical segmentation, while an 802.1Q trunk allows multiple VLANs to share a single physical connection.

---

# 💡 Question 5

## Scenario

You are configuring connectivity so that the Marketing VLAN and Engineering VLAN, which currently cannot communicate, can reach each other through the startup's single router.

## Question:

**Explain why inter-VLAN routing is required here, and justify the specific advantages of a Router-on-a-Stick configuration over adding a second physical router interface.**

## Answer:

Devices in different VLANs belong to different logical networks and cannot communicate directly at Layer 2. **Inter-VLAN routing** is therefore required to allow communication between VLAN 10 and VLAN 20.

**Router-on-a-Stick** uses one physical router interface with multiple logical sub-interfaces.

Example:

```
Router
   |
   | One physical interface
   |
802.1Q Trunk
   |
Switch
  ┌──────────┐
  │ VLAN 10  │
  │ VLAN 20  │
  └──────────┘
```

Advantages include:

1. **Cost effective** – only one physical router interface is required.
2. **Less cabling** – a single physical connection carries multiple VLANs.
3. **Scalable** – additional VLANs can be configured using additional sub-interfaces.
4. **Simple design** – VLAN routing is centrally handled by the router.
5. **Efficient use of hardware** – avoids requiring a separate physical router interface for every VLAN.

**Conclusion:**
Router-on-a-Stick is an efficient solution for a small startup because multiple VLANs can be routed through one physical router interface.

---

# 💡 Question 6

## Scenario

You are setting up a wireless access point for the startup's open-plan office, where unauthorized devices have previously connected to the network.

## Question:

**Explain the wireless security and MAC filtering options available, and justify which configuration approach best prevents unauthorized access while still allowing convenient onboarding of new employee devices.**

## Answer:

Wireless networks can use several security mechanisms, including:

- WPA2 security
- WPA3 security
- Strong wireless passwords
- MAC address filtering
- Network access controls

**MAC filtering** allows an administrator to create a list of permitted device MAC addresses. Devices whose MAC addresses are not included in the list can be prevented from accessing the wireless network.

However, MAC filtering alone should not be considered a strong security mechanism because MAC addresses can potentially be changed or spoofed.

For this startup, the recommended approach is to use **strong WPA2/WPA3 authentication with a strong password**, while MAC filtering can be used as an additional control if required.

This provides better security while allowing new employee devices to be onboarded by providing the authorized wireless credentials.

**Conclusion:**
Strong WPA2/WPA3 security should be the primary protection, while MAC filtering can provide an additional access-control layer.

---

# 💻 SECTION B — PRACTICAL CONFIGURATION TASKS

## 🛠️ Task 1: Configure Management and Staff VLANs

### 🎯 Objective

Create:

- VLAN 10 → Management
- VLAN 20 → Staff

Assign access ports to the appropriate VLANs and verify the configuration.

### Step 1: Enter privileged EXEC mode

```
enable
```

### Step 2: Enter configuration mode

```
configure terminal
```

### Step 3: Create VLAN 10

```
vlan 10
name Management
exit
```

### Step 4: Create VLAN 20

```
vlan 20
name Staff
exit
```

### Step 5: Assign Management ports

Example:

```
interface range fa0/1-5
switchport mode access
switchport access vlan 10
exit
```

### Step 6: Assign Staff ports

Example:

```
interface range fa0/6-10
```

```
switchport mode access
switchport access vlan 20
exit
```

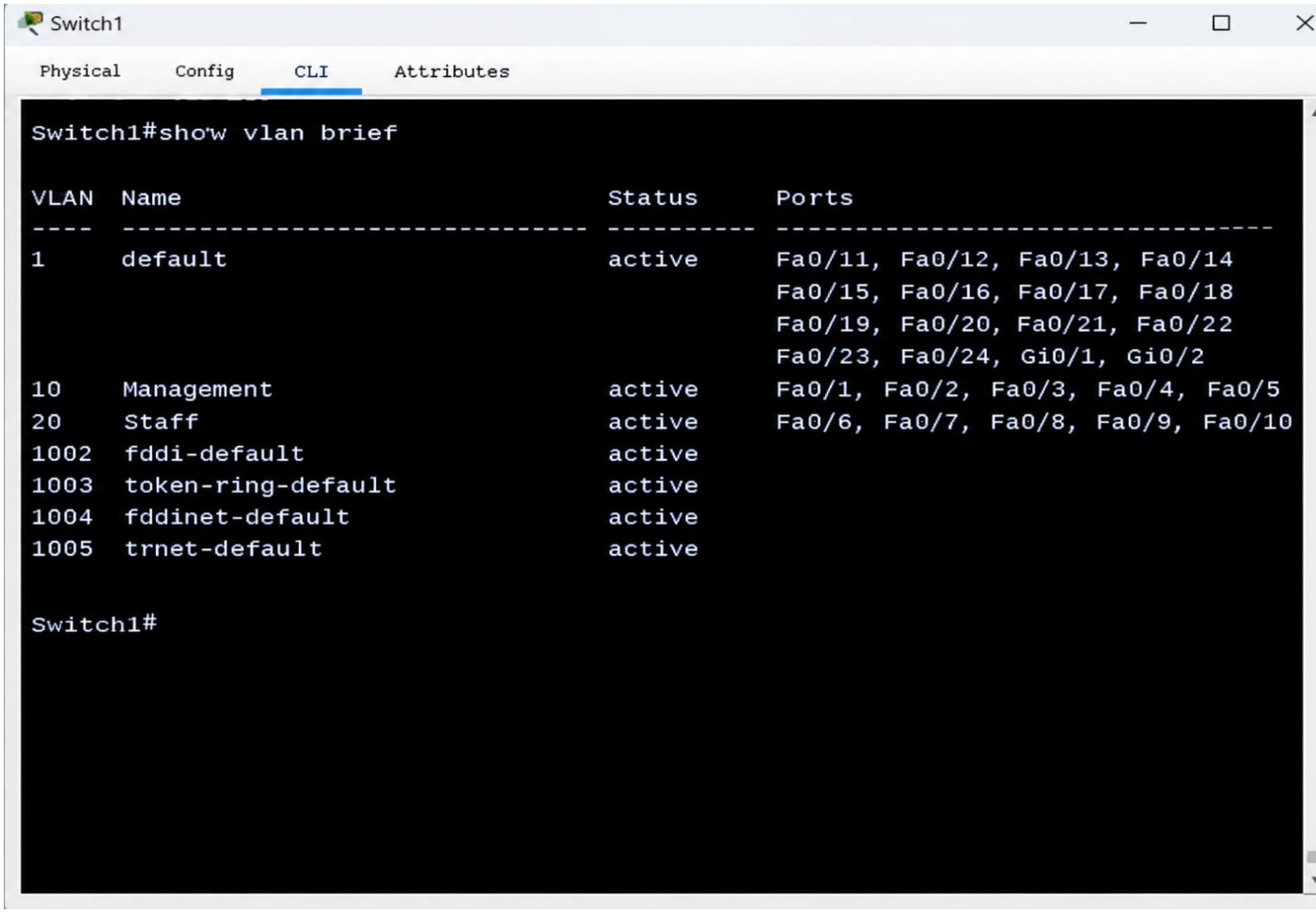
### Step 7: Verify

```
show vlan brief
```

### Expected verification

You should see VLAN 10 named **Management** and VLAN 20 named **Staff**, with the assigned ports listed under the appropriate VLAN.

**Screenshot to insert:**




---

# 🔗 Task 2: Configure an 802.1Q Trunk Link

### 🎯 Objective

Configure the switch port connected to the router as a trunk and allow only VLAN 10 and VLAN 20.

## Step 1: Select the router-connected switch interface

Example:

```
interface gigabitEthernet0/1
```

## Step 2: Configure trunk mode

```
switchport mode trunk
```

## Step 3: Restrict allowed VLANs

```
switchport trunk allowed vlan 10,20
```

## Step 4: Exit

```
exit
```

## Step 5: Verify trunk status

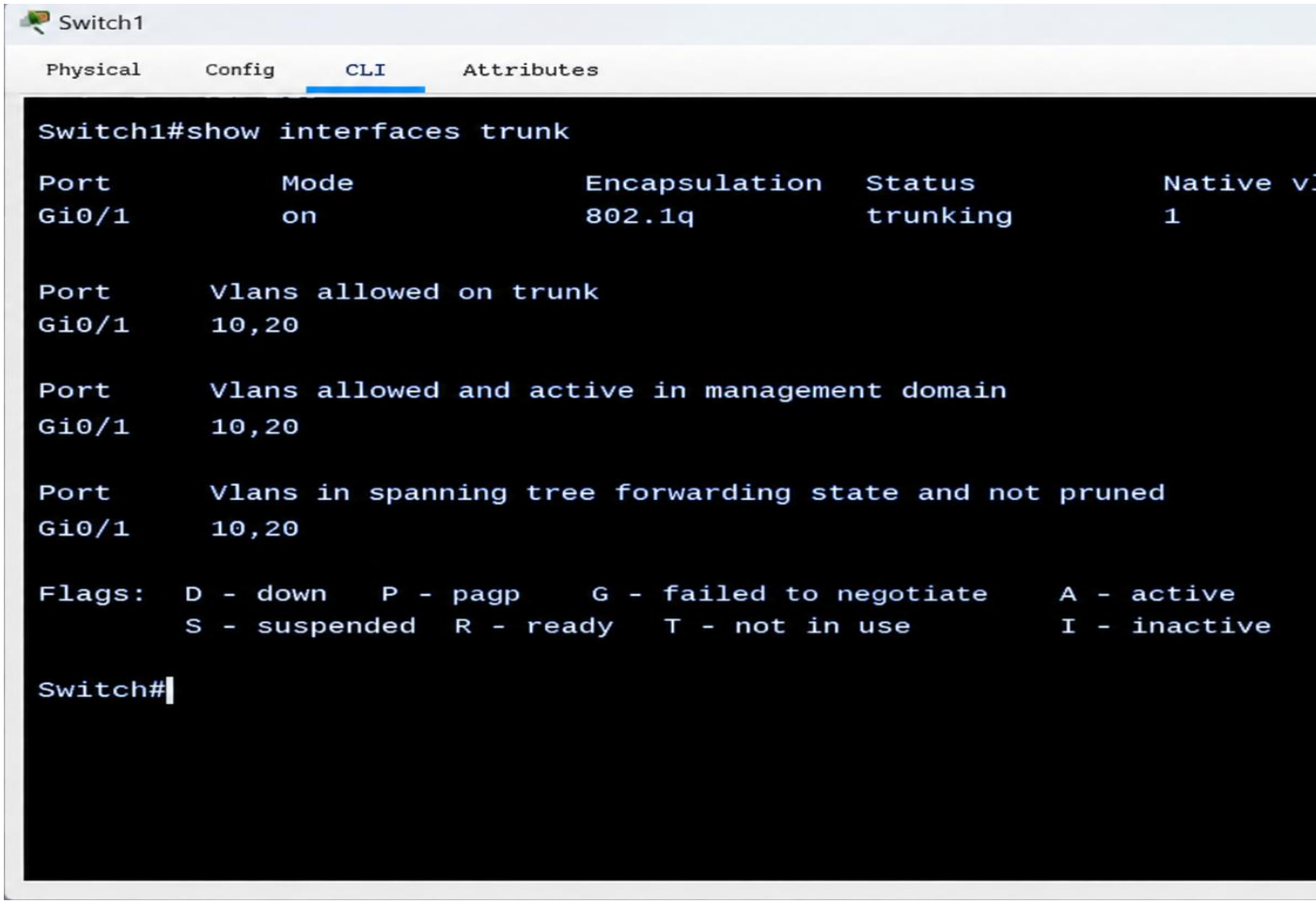
```
show interfaces trunk
```

## 🔍 Verification checklist

Confirm:

- Port is operating as a trunk.
- VLAN 10 is allowed.
- VLAN 20 is allowed.
- No unexpected VLAN is being carried.

**Screenshot to insert:**




### ⚠️ Native VLAN verification

The switch and router must use the same native VLAN configuration. A mismatch should be corrected before final testing.

---

# 🌐 Task 3: Configure Router-on-a-Stick for Inter-VLAN Routing

### 🎯 Objective

Configure router sub-interfaces for:

- VLAN 10
- VLAN 20

### Example IP addressing plan

| VLAN | Department | Gateway |
|---|---|---|
| VLAN 10 | Management | 192.168.10.1/24 |
| VLAN 20 | Staff | 192.168.20.1/24 |

## Step 1: Enter the physical router interface

Example:

```
enable
configure terminal
interface gigabitEthernet0/0
no shutdown
exit
```

## Step 2: Configure VLAN 10 sub-interface

```
interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

## Step 3: Configure VLAN 20 sub-interface

```
interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

## Step 4: Verify router interfaces

```
show ip interface brief
```

The sub-interfaces should be operational after the physical interface and trunk are correctly configured.

## Step 5: Configure host addressing

Example:

### Management PC

```
IP Address:      192.168.10.10
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.10.1
```

### Staff PC

```
IP Address:      192.168.20.10
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.20.1
```

## Step 6: Test inter-VLAN connectivity

From the Management PC:

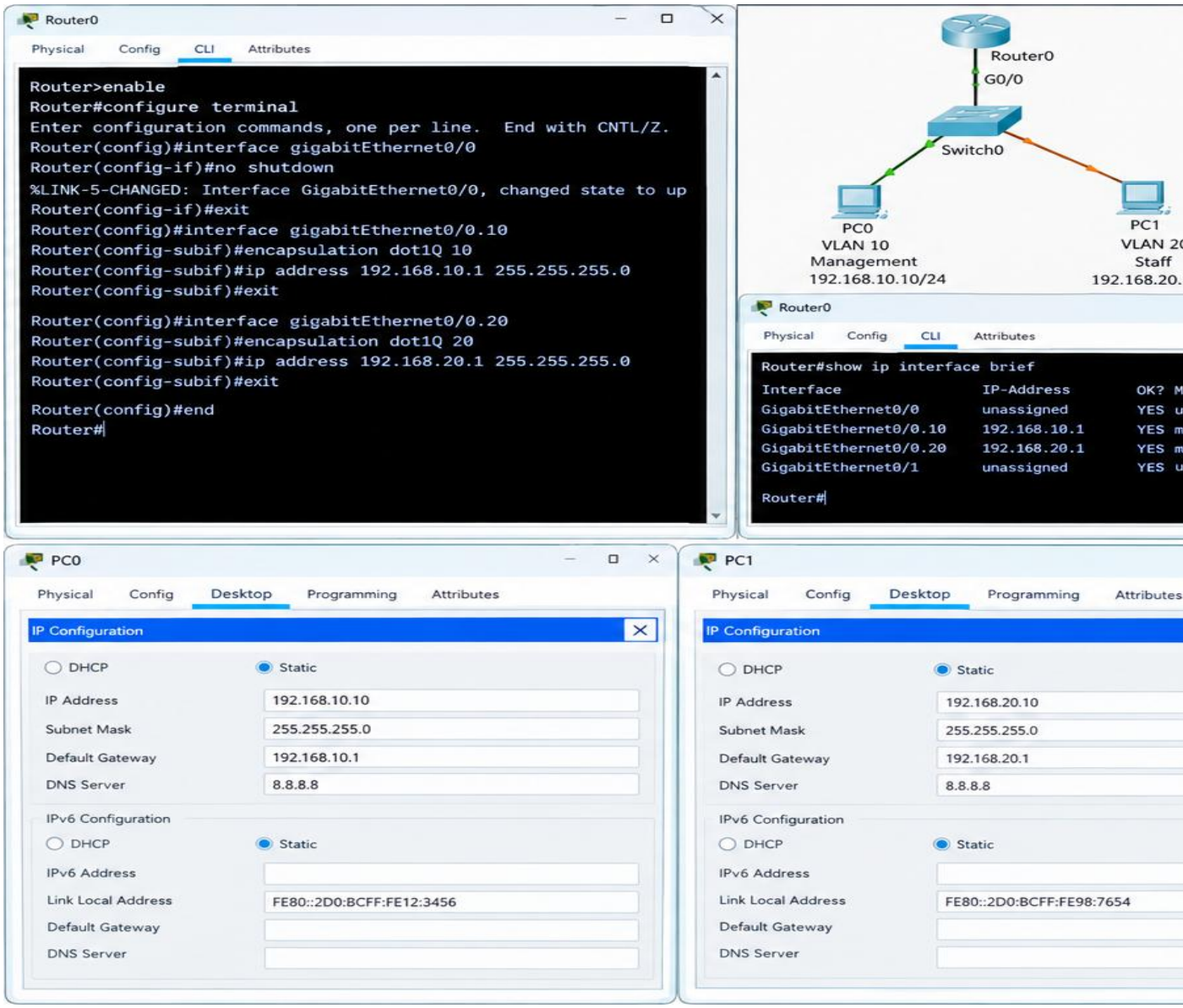
```
ping 192.168.20.10
```

### ✅ Expected result

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=...
```

Use your **actual Packet Tracer ping output** in the submission.

**Screenshot to insert:**




---

# 🔍 Task 4: Capture and Analyze Inter-VLAN Traffic

## 🎯 Objective

Capture ICMP traffic while a VLAN 10 host communicates with a VLAN 20 host.

## 📝 Procedure

1. Start the packet capture in your supported Packet Tracer/Wireshark setup.
2. Send a ping from the VLAN 10 host to the VLAN 20 host.
3. Locate the ICMP Echo Request.
4. Locate the ICMP Echo Reply.
5. Record the source and destination MAC addresses.
6. Record the source and destination IP addresses.
7. Compare the frame information before and after routing.

## 🔎 Observation

When traffic travels between different VLANs, the router performs Layer 3 routing.

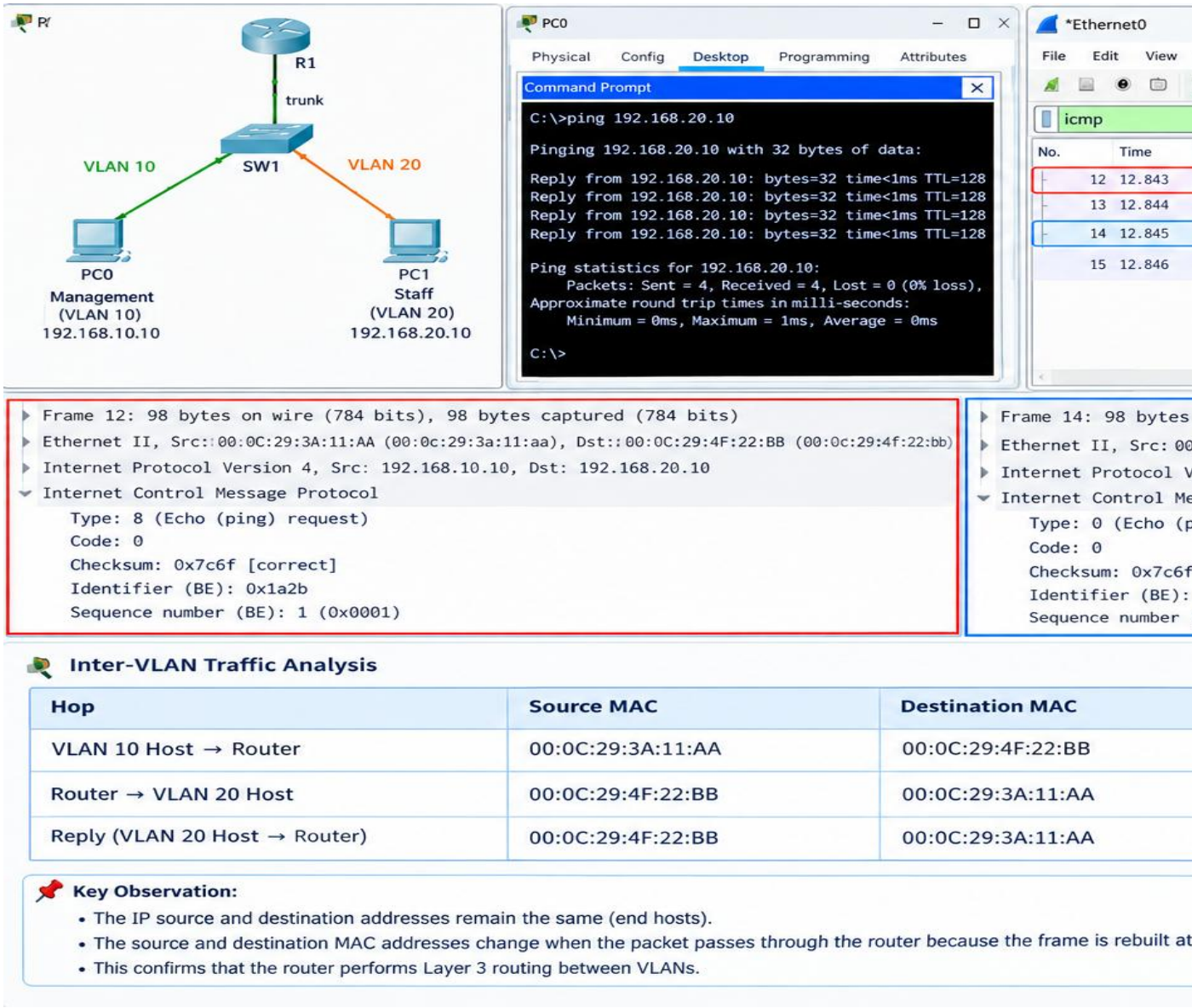
The **IP source and destination addresses represent the communicating hosts**, while the Ethernet/MAC information can change from one Layer 2 segment to another because the frame is rebuilt as it travels through the router.

## 📊 Analysis Table

| Hop | Source MAC | Destination MAC | Source IP | Destination IP |
|---|---|---|---|---|
| VLAN 10 Host → Router | [Actual value] | [Actual value] | [Actual value] | [Actual value] |
| Router → VLAN 20 Host | [Actual value] | [Actual value] | [Actual value] | [Actual value] |
| Reply | [Actual value] | [Actual value] | [Actual value] | [Actual value] |

**Important:** Enter the actual values from your capture instead of using sample addresses.

**Screenshot to insert:**




---

# 🚀 SECTION C — MINI CAPSTONE PROJECT

# 🏢 TechLaunch Startup Network Build & Fault Diagnosis

## Objective

Build a two-VLAN startup network and troubleshoot a Router-on-a-Stick failure after a simulated power outage.

The assessment requires:

- VLAN 10 → Marketing
- VLAN 20 → Engineering
- 802.1Q trunk
- Router-on-a-Stick
- Inter-VLAN routing
- Ping-based troubleshooting
- Layer 2 and Layer 3 fault isolation
- CLI verification
- Root-cause identification

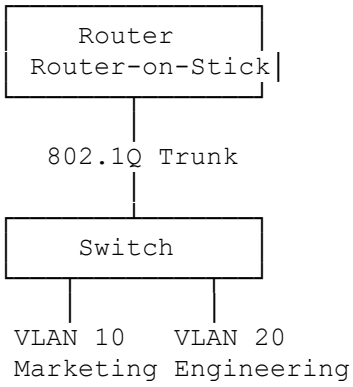
---

# 1. Network Design

## VLAN Plan

| VLAN | Department | Network | Default Gateway |
|---|---|---|---|
| 10 | Marketing | 192.168.10.0/24 | 192.168.10.1 |
| 20 | Engineering | 192.168.20.0/24 | 192.168.20.1 |

## Example topology




---

# 2. Configure VLANs

```
enable
configure terminal

vlan 10
name Marketing
exit
```

```
vlan 20
name Engineering
exit
```

Assign the appropriate access ports:

```
interface range fa0/1-5
switchport mode access
switchport access vlan 10
exit

interface range fa0/6-10
switchport mode access
switchport access vlan 20
exit
```

Verify:

```
show vlan brief
```

---

# 🔗 3. Configure the Trunk

Assuming the router is connected to `GigabitEthernet0/1`:

```
interface gigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

Verify:

```
show interfaces trunk
```

---

# 🌐 4. Configure Router-on-a-Stick

```
enable
configure terminal

interface gigabitEthernet0/0
no shutdown
exit

interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

Verify:

```
show ip interface brief
```

---

# 🛠️ 5. Fault Diagnosis

### 🚨 Problem

Marketing users can communicate with devices inside their own VLAN, but they cannot reach the Engineering file server.

The troubleshooting process should determine whether the problem is related to **Layer 2** or **Layer 3**.

---

## Step 1 — Test Local VLAN Connectivity

From a Marketing PC:

```
ping 192.168.10.1
```

### Interpretation

If the gateway responds, connectivity between the Marketing PC and its gateway is working.

---

## Step 2 — Test the Engineering Gateway

```
ping 192.168.20.1
```

If this fails while the Marketing gateway works, investigate:

- Trunk configuration
- VLAN 20
- Router sub-interface
- VLAN tagging

---

## Step 3 — Test the Engineering File Server

```
ping 192.168.20.10
```

If the router gateway is reachable but the file server is not, investigate:

- Engineering PC/server IP address
- Default gateway
- Access-port VLAN assignment
- Server availability

---

# 🔍 6. CLI Verification Commands

The following commands can be used to isolate the fault:

### Switch

```
show vlan brief
show interfaces trunk
show mac address-table
```

### Router

```
show ip interface brief
show running-config
show ip route
```

These commands help determine whether the problem is related to VLAN membership, trunking, router sub-interfaces, or Layer 3 routing.

---

# 📃 7. Root Cause and Resolution Report

### 🎯 Root Cause

**[Enter the actual fault found during your Packet Tracer troubleshooting.]**

Example:

The Engineering sub-interface was incorrectly configured after the router reboot, preventing VLAN 20 traffic from being routed correctly.

### 🛠️ Resolution

**[Enter the actual command/configuration change you made.]**

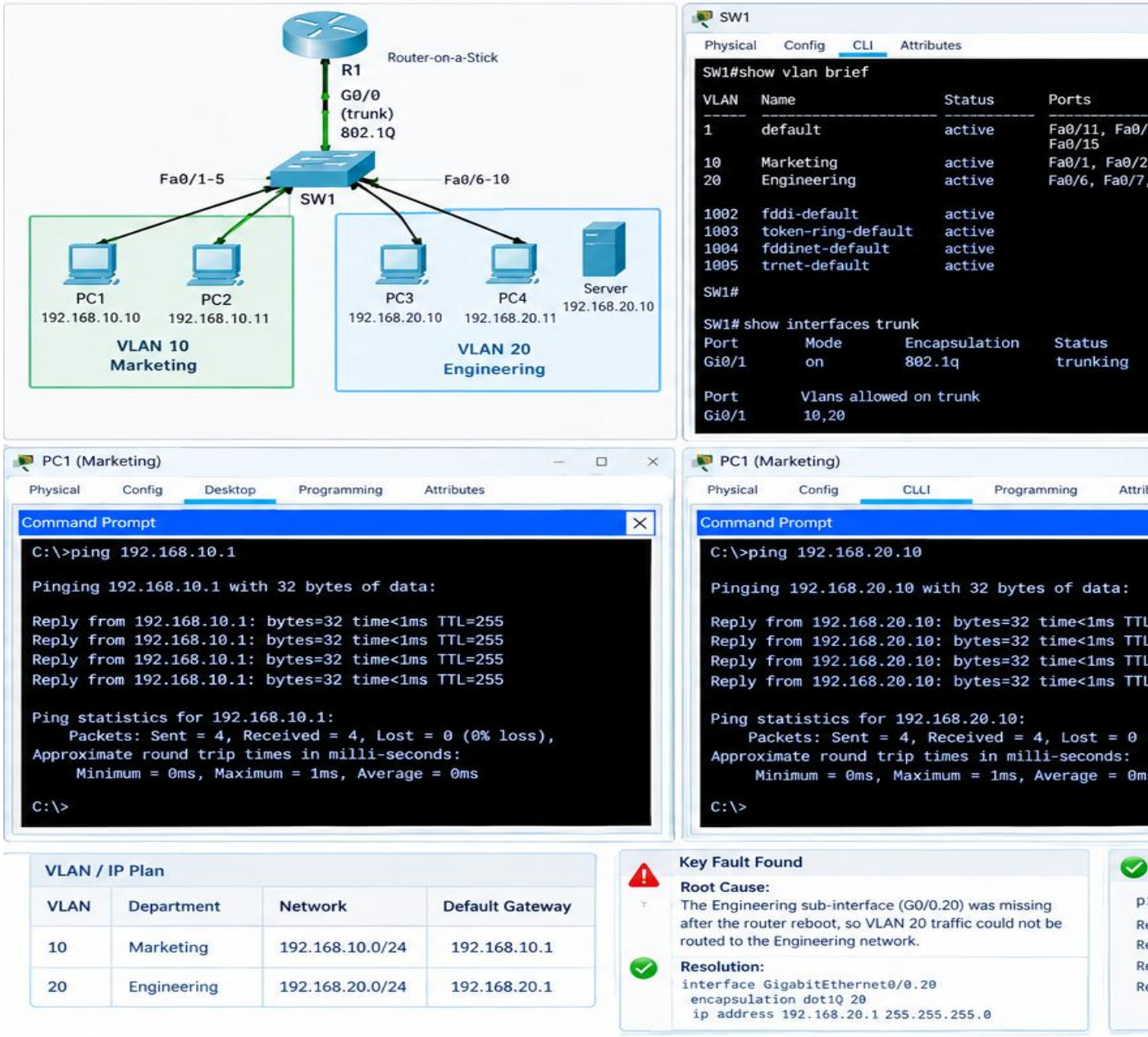
Example:

```
interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

## 🔍 Verification

```
ping 192.168.20.10
```

**Screenshot to insert:**




---

# 🤖 SECTION D — AI-AUGMENTED LEARNING

## 🤖 Step 1 — Build With AI

The assessment requires an AI tool to help generate a complete Cisco IOS configuration for:

1. VLAN 10
2. VLAN 20
3. Port assignments
4. 802.1Q trunk
5. Router-on-a-Stick
6. Inter-VLAN connectivity

The assessment specifically requires submission of the **exact prompt given to the AI tool**.

### 💬 AI Prompt Used

Create a complete Cisco Packet Tracer configuration for a small startup network using VLAN 10 and VLAN 20. Configure VLAN 10 and VLAN 20 on a Cisco switch with appropriate access-port assignments. Configure an 802.1Q trunk between the switch and router, allowing only VLAN 10 and VLAN 20. Configure Router-on-a-Stick sub-interfaces on the router with appropriate IP addresses and VLAN encapsulation. Provide the complete Cisco IOS commands in the correct order and include verification commands such as show vlan brief, show interfaces trunk, show ip interface brief, and ping testing between a VLAN 10 host and a VLAN 20 host.

---

# 💻 AI-Generated Configuration Script

```
! SWITCH CONFIGURATION

enable
configure terminal

vlan 10
name Management
exit

vlan 20
name Staff
exit

interface range fa0/1-5
switchport mode access
switchport access vlan 10
exit

interface range fa0/6-10
switchport mode access
switchport access vlan 20
exit

interface gigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

```
end
```

### Router Configuration

```
enable
configure terminal

interface gigabitEthernet0/0
no shutdown
exit

interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

end
```

### 🔍 Verification Commands

```
show vlan brief
show interfaces trunk
show ip interface brief
```

### 🧪 Ping Test

From the VLAN 10 host:

```
ping 192.168.20.10
```

---

# 🧪 Step 2 — Test & Debug Without AI

## 🐞 Bug Identified

**bug investigate:**

The router's physical interface may not be enabled after the configuration is applied.

### 🛠️ Correction

```
interface gigabitEthernet0/0
no shutdown
```

### 💡 Why This Was Required

The router's physical interface must be operational for the configured sub-interfaces to communicate with the switch through the trunk.

---

# 🔄 AI Configuration vs Corrected Configuration

| Item | AI Version | Corrected Version |
|---|---|---|
| VLAN 10 | Configured | Verified |
| VLAN 20 | Configured | Verified |
| Access ports | Configured | Verified |
| Trunk | Configured | Verified |
| Allowed VLANs | 10,20 | 10,20 |
| Router sub-interface | Configured | Verified |
| Physical interface | **Potential missing `no shutdown`** | `no shutdown` added |
| Inter-VLAN ping | Tested | Re-tested after correction |

---

# 📝 3–4 Line Explanation

During testing, I identified a configuration issue related to the router interface. The physical router interface needed to be enabled using the `no shutdown` command so that the Router-on-a-Stick sub-interfaces could operate correctly. After applying the correction, I verified the interface status and performed an inter-VLAN ping test. The corrected configuration successfully restored communication between the two VLANs.

---

# ✅ FINAL VERIFICATION CHECKLIST

| Verification | Command/Test | Status |
|---|---|---|
| VLAN 10 created | `show vlan brief` | ☐ |
| VLAN 20 created | `show vlan brief` | ☐ |
| Access ports assigned | `show vlan brief` | ☐ |
| Trunk configured | `show interfaces trunk` | ☐ |
| VLAN 10 allowed | `show interfaces trunk` | ☐ |
| VLAN 20 allowed | `show interfaces trunk` | ☐ |
| Router interface up | `show ip interface brief` | ☐ |
| Sub-interface VLAN 10 | `show ip interface brief` | ☐ |

| | | |
|---|---|---|
| Sub-interface VLAN 20 | `show ip interface brief` | ☐ |
| Inter-VLAN connectivity | `ping` | ☐ |
| Wireshark/traffic analysis | Packet capture | ☐ |
| Capstone fault identified | CLI + ping testing | ☐ |
| AI bug identified | Independent testing | ☐ |
| AI correction tested | Final ping | ☐ |
| `.pkt` file saved | Packet Tracer | ☐ |
| Screenshots attached | Evidence | ☐ |

---

# 🏁 CONCLUSION

This assessment demonstrates the practical application of fundamental **Routing and Switching** concepts. The activities cover static and dynamic routing, RIP and OSPF, MAC address learning, VLAN segmentation, 802.1Q trunking, Router-on-a-Stick, inter-VLAN routing, traffic analysis, troubleshooting, and AI-assisted configuration.

The practical tasks demonstrate how VLANs can logically separate departments while trunking and Router-on-a-Stick provide communication between different VLANs. The troubleshooting section further demonstrates a structured approach to identifying Layer 2 and Layer 3 connectivity problems.

Finally, the AI-augmented section demonstrates that AI-generated configurations should always be independently tested, verified, and corrected when necessary.